Designing a Gamified Application for User Privacy Preference Extraction

Continual Assessment Report

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Abstract

[Write this after the main sections are drafted.]

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Chapter 1

Introduction

1.1 Background

Even in this day and age of absurd technological advancements, user privacy preference remains an alien topic for the general public. There are several ways user privacy preferences can be collected online, some common ways that are used may be through surveys [1], device settings parsing [2], and cookies' consent banner interaction [3]. This project will specifically be exploring how user privacy preferences are extracted through cookies, commonly presented to users with the interface of a popup provided by CMPs (Consent Management Platforms) called cookie banners [4] each time they visit a website. These cookie banners would then request permissions from users to indicate their consent preferences for cookies, which are small blocks of data placed on the user's device that allows the websites to store stateful information regarding the user or the user's browsing activity [5].

Most people don't really pay close attention to the user privacy preferences that they (have to) set each time they visit a new website on the Internet and is presented with a cookie banner. Whenever people are presented with a popup-like interface with verbose texts that they have to read through, the instinct for most people would be to get rid of the popup window [6]. As such when combined with the assumption that the general public isn't sufficiently educated on the topic of the importance of user privacy preferences and how to manage them well, users tend to click the options presented on the interface that allows them to proceed immediately (i.e. usually the "Accept All" or "Deny All" options). Some reasons why people tend to ignore how user privacy preference extraction works in cookie banners is likely driven by the cookie banners' deliberately complex design and users' knowledge limitations [7].

In view of this problem presented and potential causes that are established, this project aims to solve this issue by gamifying the process of user privacy preference extraction through cookie banners.

1.2 Project Objectives

Given the background of the project, the project aims to address the problems through primarily 3 objectives:

1. Investigate existing solutions that attempts to address similar issues
2. Design a solution that presents the management of cookies in a gamified interface
3. Evaluate how a gamified process may be effective in helping users manage their privacy preferences within their own initiative

With these 3 objectives as the foundational directive pointers for this report, the report will be detailing how the solution for this project came to ideation, the project's progress so far and it's subsequent analysis on how effective it is in achieving the objectives.

1.3 Ideation of Solution

Before constructing the ideation of this project's solution, it is useful to understand where the current common practices stands, specifically how do most websites collect user's cookies consent preferences.

1.3.1 Review of Current Cookie Consent Practices

As mentioned briefly in the introduction of the report, most websites currently uses specialized tools called Consent Management Platforms (CMPs) to help them log user's consensual choices for cookies that their websites will be using [8]. In the form of popups, user will be asked to indicate their preferences whenever they visit a website that their browser does not have cached data recording their privacy preference choices for the cookies.

Exhibit A: F1 Official Website

We will be taking the F1 official website [9] as an example of how this looks practically. Upon entering the website for the first time, users are greeted with the popup window shown in Figure 1.1.

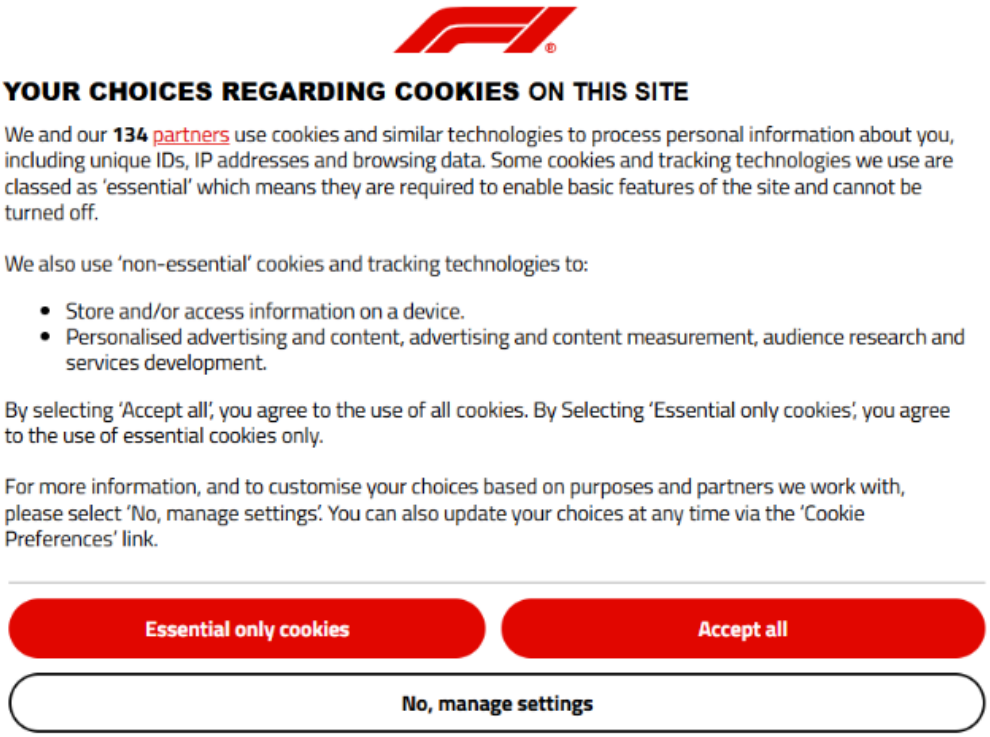


Figure 1.1: Cookie consent banner displayed on the F1 official website.

Users are presented with mainly 3 choices on the interface, one to accept only essential cookies, one to accept all, and the last one is to have more autonomy and manually manage settings for cookie consent choices. Observe how even though details regarding the cookies and what they are mainly used for is explicitly written on the interface, it is presented in a way that may seem overly verbose and irrelevant for the common public to understand. In addition, the options that are highlighted in red will also appeal more towards the user, which causes users to make choices that subject themselves to consent more cookies. This is a common practice among most websites known as dark patterns, that attempt to use manipulative UX designs to force user into making choices that often benefits the company behind [10].

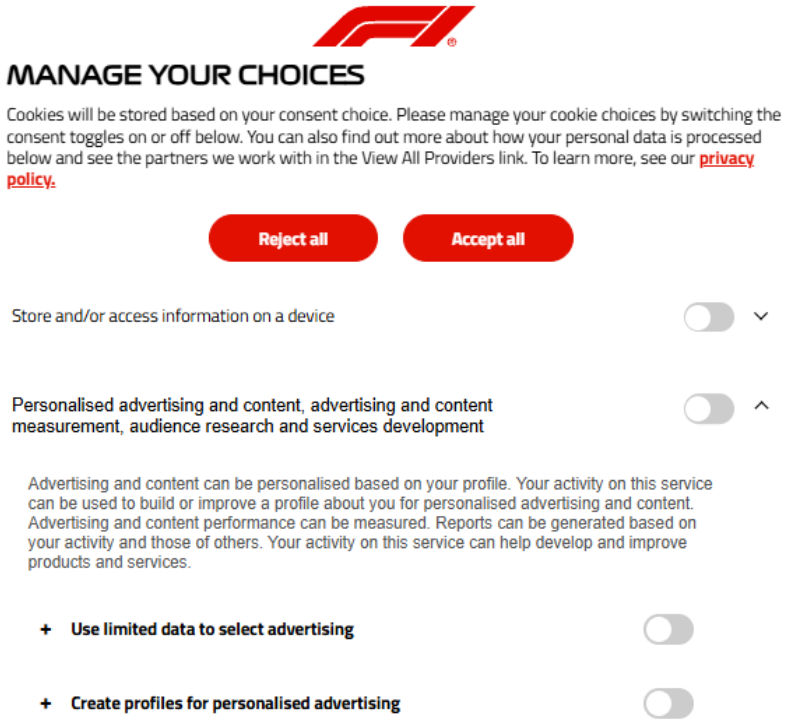


Figure 1.2: Cookie banner 'manage settings' window of F1 official website.

Upon clicking the 'No, manage settings' option, user can further indicate their preference for which purpose their collected data from the cookies can be used. The usage of cookies are split into different ways data collected through cookies can be used, such as to select advertising, or to create profiles for personalised advertising, as shown in Figure 1.2. Though the user is able to have more autonomy, this also adds to the problem of a complex privacy preference extraction process as user has to go through each options to decide whether they want to consent to it or not.

Exhibit B: Fifa Official Website

Another example that we can look at is from the Fifa Official Website [11]. Similarly, they have a cookie banner for the user to indicate their privacy preference through 3 buttons as shown in Figure 1.3.



Figure 1.3: Cookie consent banner displayed on the Fifa official website.

If the user were to click the preference center option marked in white, the user would then be greeted with another interface as such under Figure 1.4. Cookies that can be consented such as functional cookies are all compiled into one window in the privacy preference center interface. This may seem easier for the users to customize their privacy

preference, it took away some autonomy from the user when they can't consent at a deeper level where they consent only to certain functional cookies instead of all functional cookies as a whole, comparatively to F1's cookie banner in Figure 1.2 where user is able to indicate preference for each cookies' purpose. However, this is significantly more concise and easy for the user to go through.

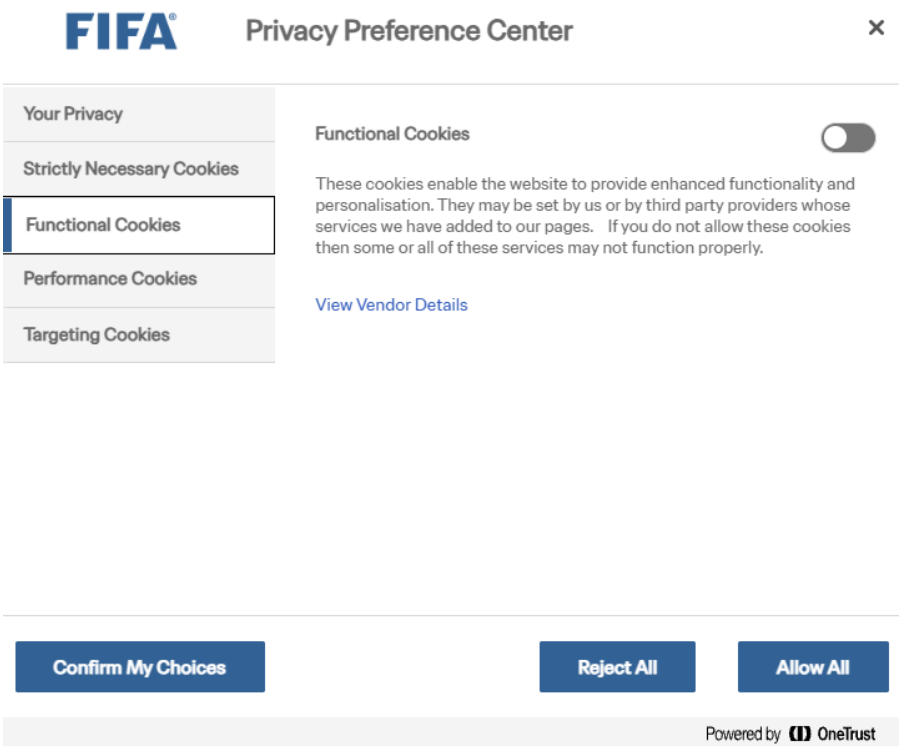


Figure 1.4: Privacy Preference Center of Fifa official website.

1.3.2 Requirements

After reviewing the common cookie consent preference extraction practices of websites, we can come up with 5 requirements of how our solution should look like in order to solve the problems found above.

1. Concise

   Users shouldn't be looking at verbose content to understand the implication of their privacy preference choices, confusion should be avoided

2. Convenient

   User should be able to easily access and set up the application

3. Engaging

   If there are a lot of privacy preferences to be extracted, users should be engaged when doing so

4. Informative

   Provides sufficient information to the user on how their privacy preferences affects them

5. User-centric autonomous

   Users should be able to have as much control as possible to their own cookie consent privacy preference

1.3.3 Constructing the Solution

A cozy/pixel-art privacy companion that appears when you visit websites, explains what privacy choices are being requested, and turns privacy decisions into an interactive scenario-based experience.

Notion

User privacy preference is a subject that each user passes by everyday as they visit different websites, yet most people ignore it. There are multiple reasons as to why this happens, but the most prominent and straightforward reason that users tend to overlook it is that the current way of presenting it is boring, verbose and inconvenient, encountering dozens of such popups would be mentally tiring for users [7]. Hence, the topic of designing a gamified application to represent the simulation of managing each user's own privacy preference aims to solve the issue of the current method of extracting these preferences from users. The intrinsic details of user privacy preferences can be simplified and become more approachable for users. A visual representation of everything is useful to engage the user while presenting each cookie, third-parties, privacy preferences etc. in a new way that catches the attention of the user.

Use of analogy in scenario-based questions

As browsing the internet and different websites is just like visiting different people's homes, instead of presenting websites and cookies as technical terms, websites can be described as homes that users visit and the cookies are like guests in the home. Subsequently, they can be personified according to the purpose of each cookies. For example, authentication cookies that let websites recognizes users can be presented as the owner (company) of the home (website) wants to remember the user's name so that the user can re-enter their home anytime [12].

Addressing the Requirements

1.3.4 Constraints

Chapter 2

Literature Review

2.1 Review approach

[Describe where and how you searched, search terms, date range, and how you selected sources.]

2.2 Key themes

2.2.1 [Theme 1]

[Compare and synthesize relevant studies; do not just summarize one paper per paragraph.]

2.2.2 [Theme 2]

[Explain agreements, differences, limitations, and what remains unresolved.]

2.3 Research gap and implications for this project

[Show how the reviewed work motivates your objectives and design choices.]

Chapter 3

Progress to Date

3.1 Work completed

[Describe completed investigation, requirements, implementation, or other evidence of progress.]

3.2 Design or investigation alternatives

[Present plausible alternatives and compare them against clear criteria. A table can help.]

Table 3.1: Example comparison of alternatives (replace with your own).

Criterion	Alternative A	Alternative B
[Criterion]	[Evidence / trade-off]	[Evidence / trade-off]

3.3 Preliminary design or investigation

[Describe the current preferred direction, why it was selected, assumptions, and open questions.]

3.4 Challenges and risks

[Explain issues encountered and what you did or will do about them.]

Chapter 4

Research Plan for Next Semester

4.1 Planned work and methods

[List the remaining activities, methods, tools, and expected outputs.]

4.2 Milestones

Table 4.1: Provisional next-semester plan. Replace weeks with actual dates.

Period	Activity	Output
Weeks 1–2	[Refine design / protocol]	[Approved plan]
Weeks 3–5	[Build / collect / investigate]	[Intermediate result]
Weeks 6–8	[Evaluate / analyze]	[Evaluation findings]
Weeks 9–12	[Refine and write up]	[Final report / demonstration]

4.3 Evaluation and success criteria

[State how you will judge whether objectives have been met and what evidence you will collect.]

Chapter 5

Conclusion

[Briefly recap the problem, progress, current direction, and immediate next steps.]

References

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